

The impact of tropospheric versus stratospheric CO₂ forcing on the Hadley cell

Climate dynamics department meeting

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June 26, 2026



MAX-PLANCK-INSTITUT
FÜR METEOROLOGIE



Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG

1 Introduction

2 Held&Hou theory

3 Incorporating eddies

Changes in Hadley circulation

	Uniform	Tropo	Strato
$\Delta\psi_{\max,\text{NH}}$	-1.6%	3.4%	-8.7%
$\Delta\psi_{\max,\text{SH}}$	-1.4%	5.6%	-5.7%
$\Delta\Phi_{\text{H,NH}}$	1.6°	0.8°	0.8°
$\Delta\Phi_{\text{H,SH}}$	-1.6°	-1.0°	-0.6°

Surprise

- HC is **weakening** with stratospheric CO₂
- HC is **strengthening** with tropospheric CO₂

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Linearisation of direct thermodynamical drivers

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{5}{2} \frac{\delta H}{H} + \frac{5}{2} \frac{\delta\Delta_h}{\Delta_h} - \frac{\delta\Delta_v}{\Delta_v}$$
$$\frac{\delta\Phi_H}{\Phi_H} = \frac{1}{2} \frac{\delta H}{H} + \frac{1}{2} \frac{\delta\Delta_h}{\Delta_h}$$

- meridional near-surface potential temperature contrast

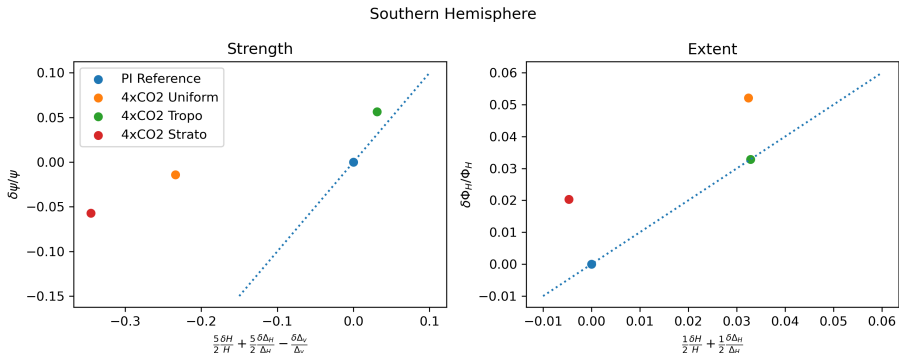
$$\Delta_h = \bar{\theta}(800 - 700 \text{ hPa}, 0 - 10^\circ) - \bar{\theta}(800 - 700 \text{ hPa}, 80 - 90^\circ)$$

- subtropical static stability

$$\Delta_v = \bar{\theta}(200 \text{ hPa}, 20 - 40^\circ) - \bar{\theta}(800 \text{ hPa}, 20 - 40^\circ)$$

- tropical tropopause height $H = \bar{H}(0 - 15^\circ)$

Does linearised H&H explain the HC's response?



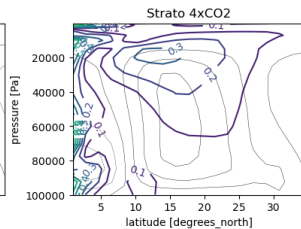
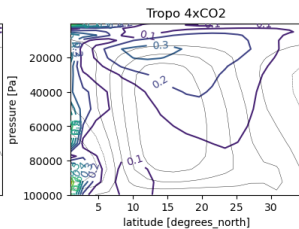
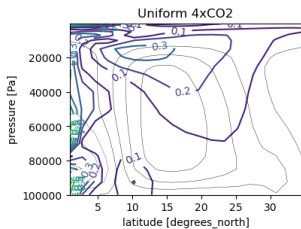
Why does H&H fail to explain HC dynamics?

Limitations

- Idealized model without oceans, land mass, clouds, etc.
- Eddies and axially asymmetric flow
- Temperature scaling parameters are in radiative equilibrium

- 1 Introduction
- 2 Held&Hou theory
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Local Ro number



Eddy dominant regime

Eddy momentum flux

$$f\bar{v} = \frac{1}{a \cos^2 \varphi} \frac{\partial(\cos^2 \varphi \overline{u'v'})}{\partial \varphi} + \frac{\partial \overline{u'\omega'}}{\partial p} \quad (1)$$

$$\begin{aligned} \psi_{\max} &= \frac{2\pi a \cos \varphi}{g} \int_{p_{\max}}^0 \bar{v} dp' \\ &= \frac{2\pi a \cos \varphi}{fg} \int_{p_{\max}}^0 \left(\frac{1}{a \cos^2 \varphi} \frac{\partial(\cos^2 \varphi \overline{u'v'})}{\partial \varphi} + \frac{\partial \overline{u'\omega'}}{\partial p} \right) dp' \end{aligned}$$

HC strength and eddy momentum fluxes

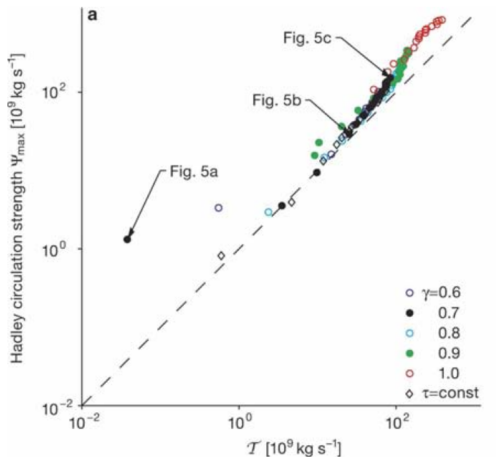
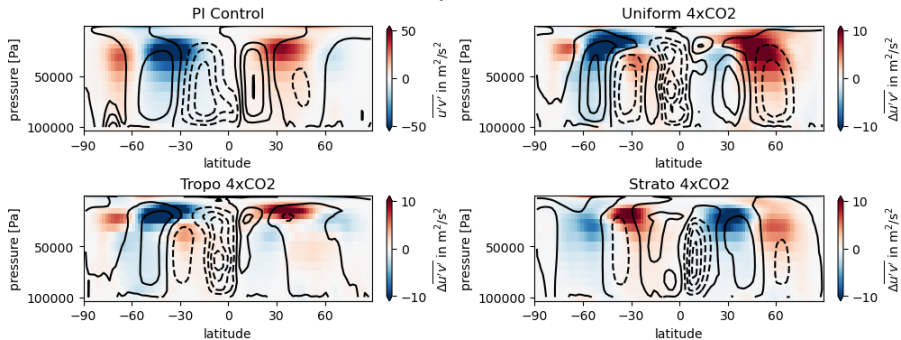
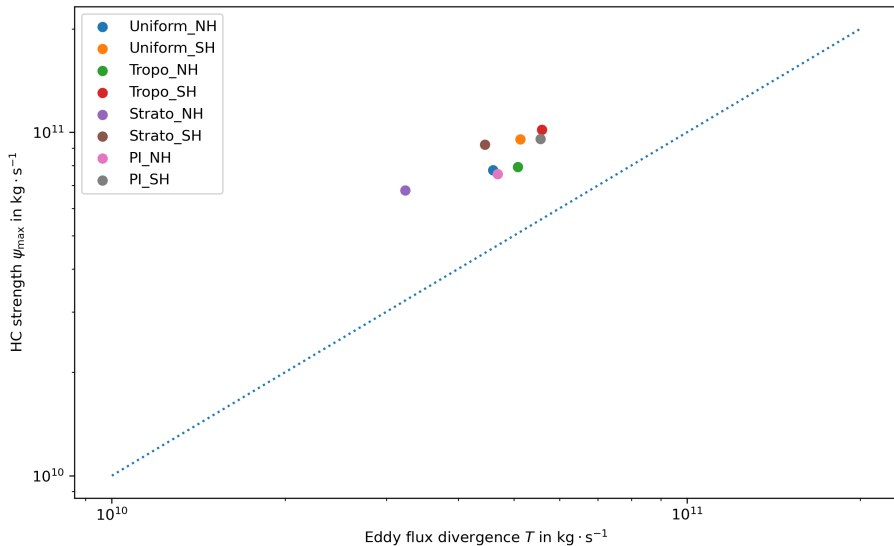


Figure: Linear relation between eddy momentum flux and HC strength from [Walker and Schneider, 2006].

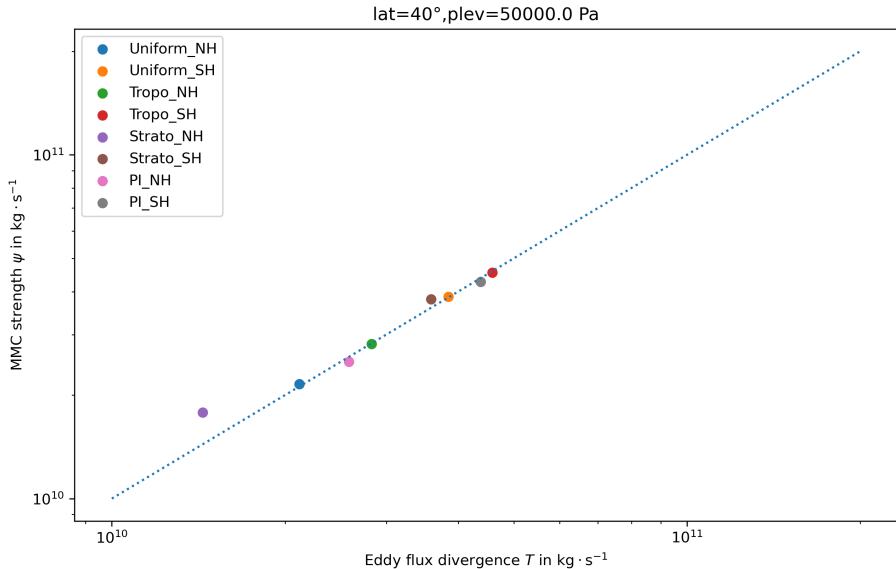
Anomalous eddy momentum fluxes



HC strength and eddy momentum fluxes



Eddy momentum fluxes and MMC in mid latitudes



Energetic framework

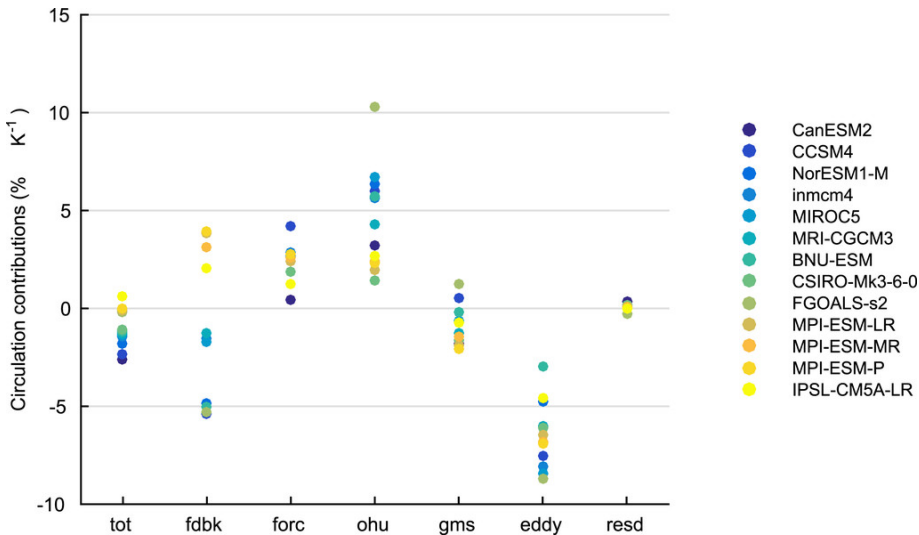
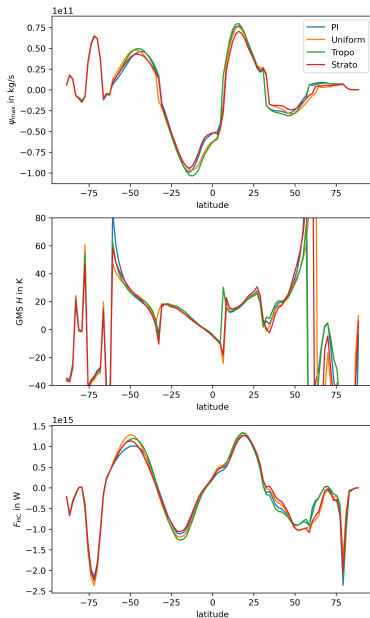


Figure: From [Feldl and Bordoni, 2016]

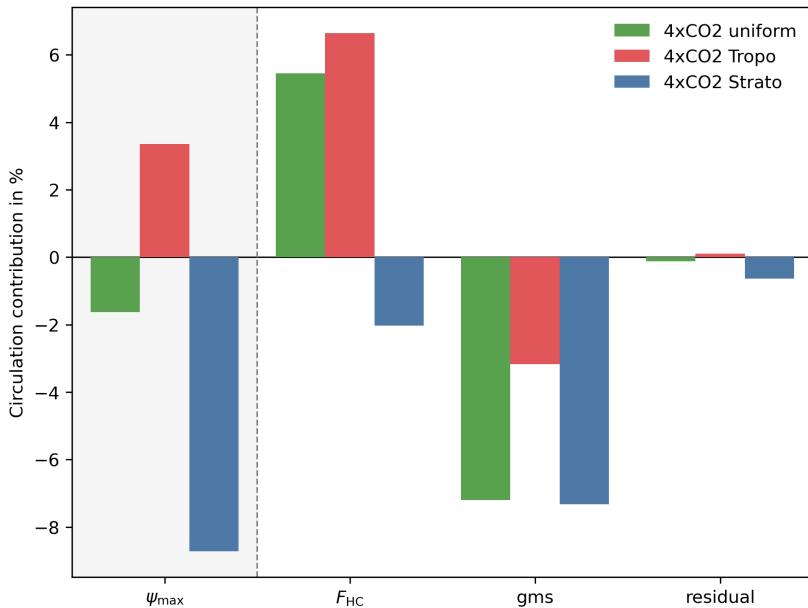
Energy transport by the Hadley cell

$$\begin{aligned} F_{\text{HC}} &= \psi_{\text{max}} H \\ &= \frac{2\pi a \cos \phi}{g} \int_{p_0}^0 [\bar{h}][\bar{v}] dp \\ \longrightarrow \frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} &= \frac{\delta F_{\text{HC}}}{F_{\text{HC}}} - \frac{\delta H}{H} \end{aligned}$$

Energy transport by the Hadley cell



Contributions



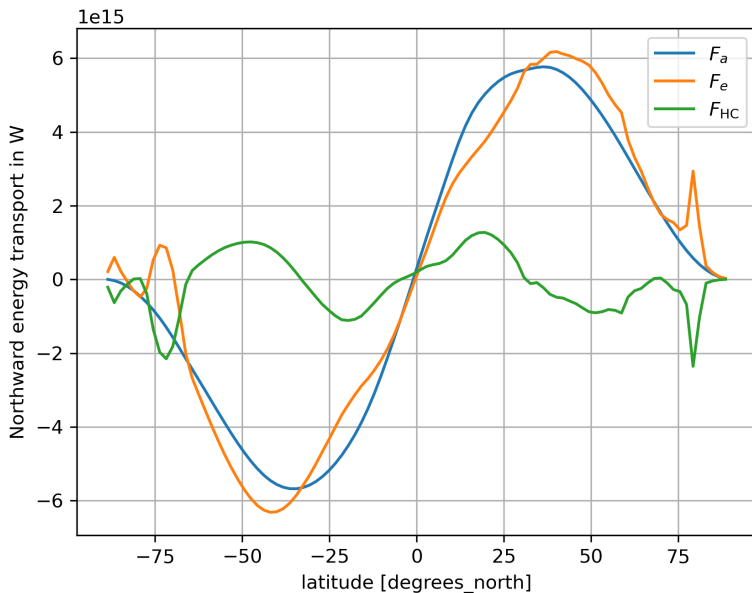
Total atmospheric energy transport

$$F_a = \int_{-\pi/2}^{\phi} \int_0^{2\pi} R_{\text{TOA}} a^2 \cos \phi d\phi d\lambda = F_{\text{HC}} + F_e$$

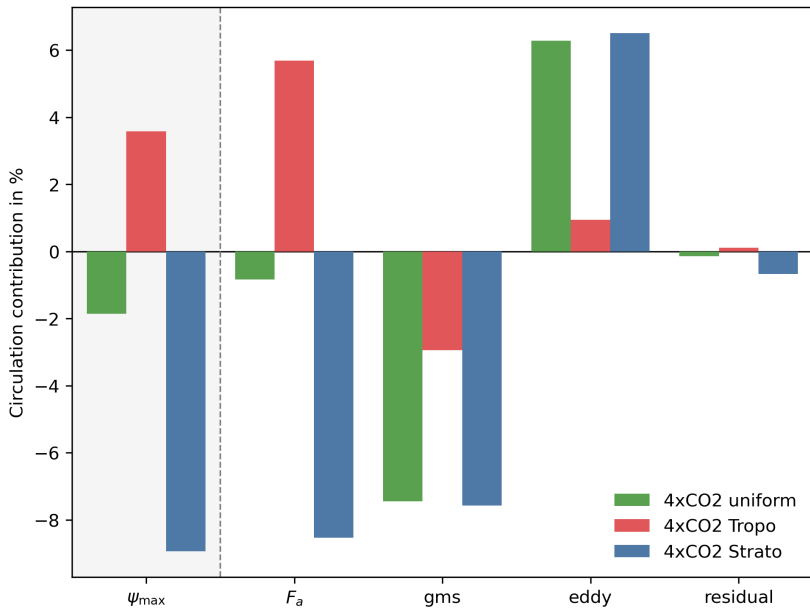
$$R_{\text{TOA}} = \text{SW}_{\text{net}} + \text{OLR}$$

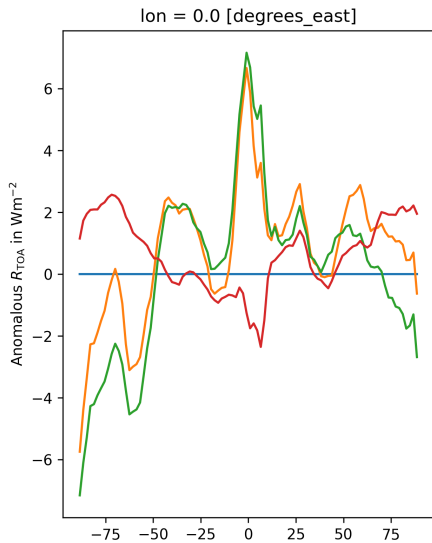
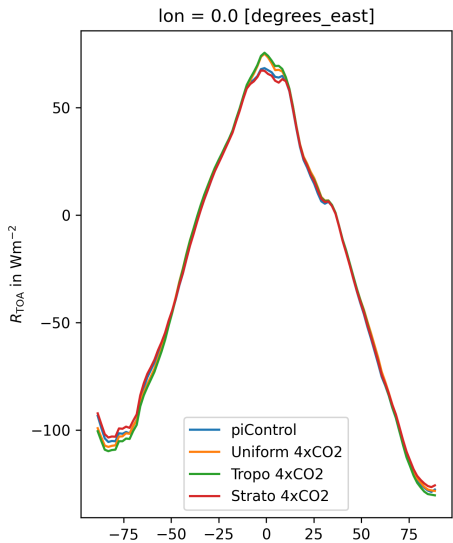
$$\rightarrow \frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_a - \delta F_e}{F_{\text{HC}}} - \frac{\delta H}{H}$$

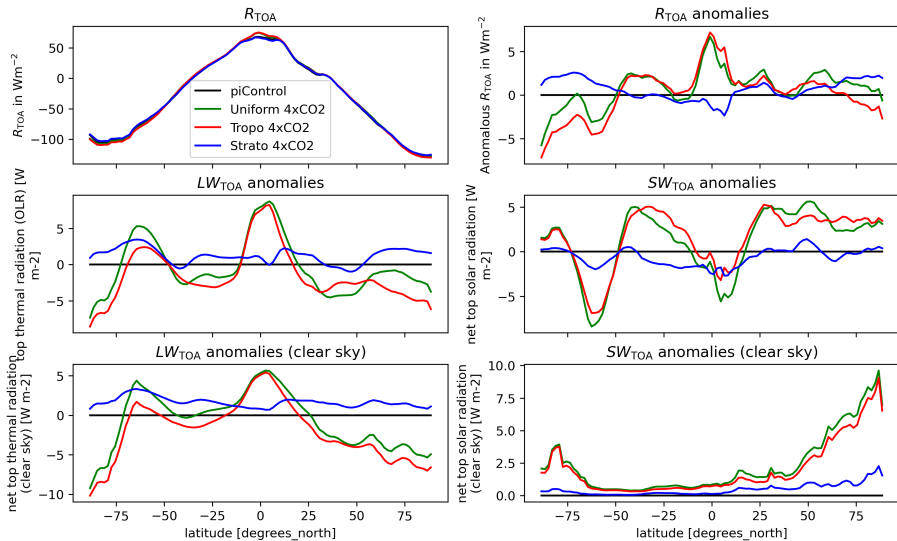
Total atmospheric energy transport



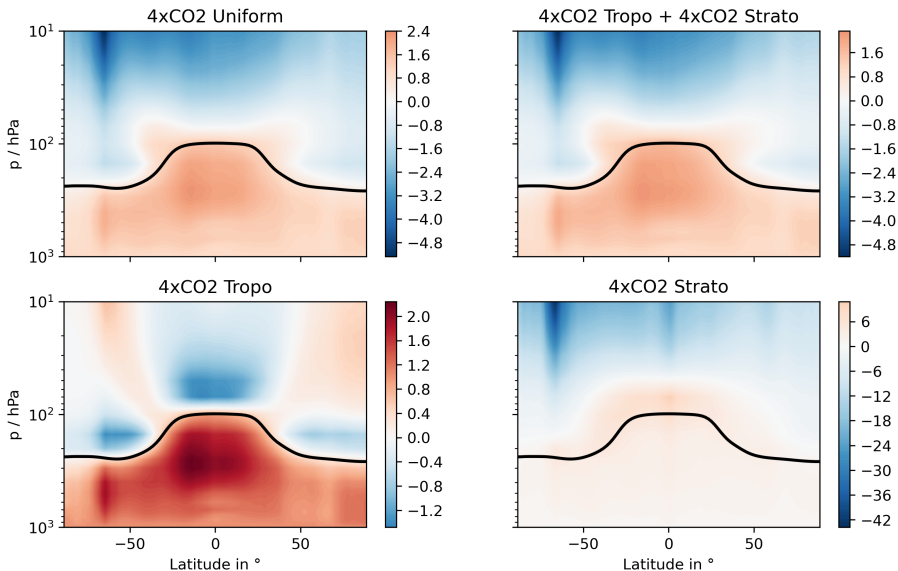
Contributions



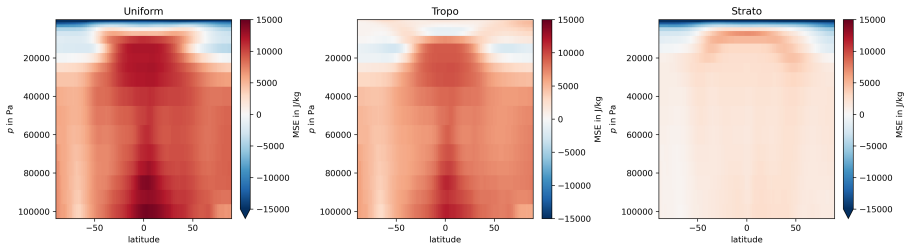




Temperature anomalies normalised



MSE anomalies



References I

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[Levine and Schneider, 2011] Levine, X. J. and Schneider, T. (2011).

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[Vallis, 2017] Vallis, G. K. (2017).

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Cambridge University Press, 2 edition.

[Walker and Schneider, 2006] Walker, C. C. and Schneider, T. (2006).

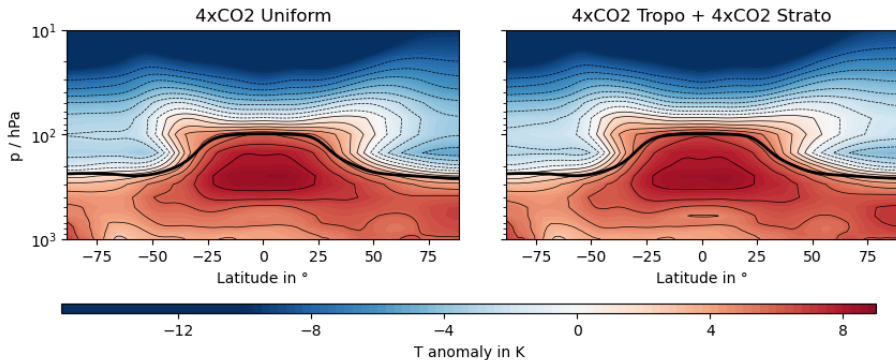
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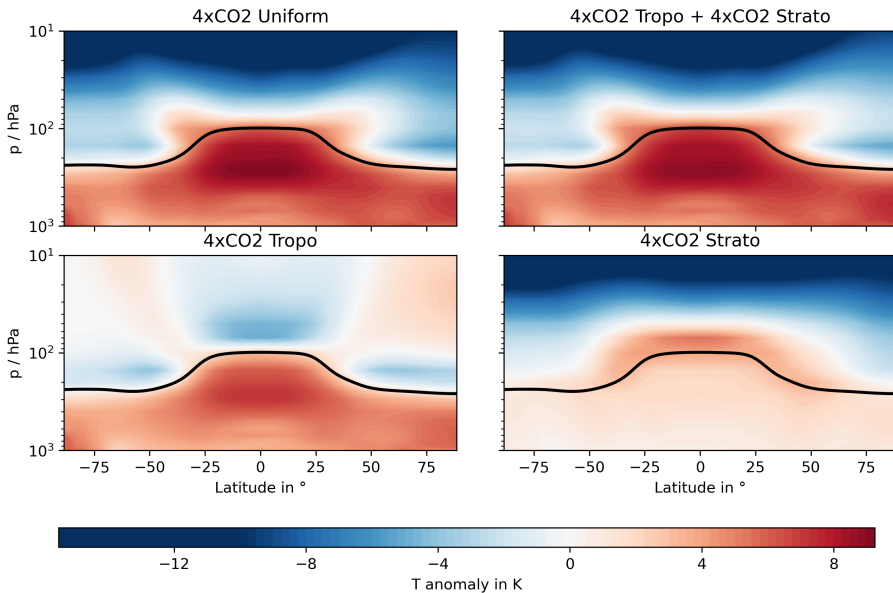
Model: ECHAM 6

- CMIP6 type general circulation model
- Mixed-layer ocean ($d = 50$ m) configuration
- 47 vertical pressure levels with (lat,lon)= (96, 192) gridpoints (~ 200 km)
- 4 runs: pre-industrial, uniform $4\times\text{CO}_2$, $4\times\text{CO}_2$ troposphere, $4\times\text{CO}_2$ stratosphere

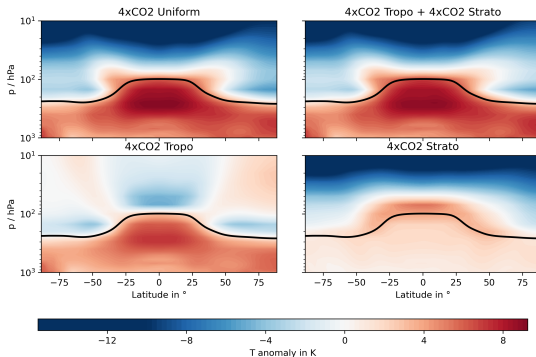
Linear decomposition of forcing works



Temperature anomalies



Temperature anomalies



Yet to be clarified

- Stratospheric cooling in 4xCO₂ Tropo
- Stratospheric warming in 4xCO₂ Strato
- one-dimensional RCE model *konrad*

The axisymmetric theory of the HC

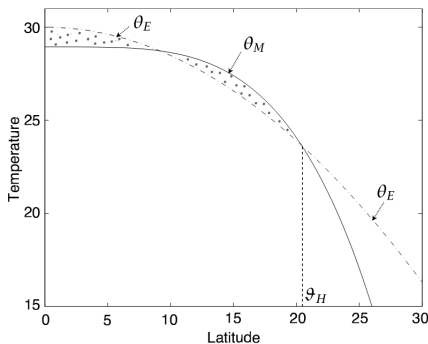


Figure: The edge of the HC [Vallis, 2017].

Width and extent of the HC [Held and Hou, 1980]

$$\Phi_H \propto (H\Delta_h)^{1/2} \quad (2)$$

$$\psi_{\max} \propto \frac{(H\Delta_h)^{5/2}}{\Delta_v} \quad (3)$$

H : Troposphere depth

Δ_h : Fractional pole to equator temperature difference

Δ_v : Fractional surface to tropopause temperature difference

Indirect thermodynamical drivers

Eddy momentum flux

$$\psi(1 - \text{Ro}) \approx \frac{S}{f} \quad (4)$$

- $\text{Ro} \rightarrow 1$: Axisymmetric limit
- $\text{Ro} \rightarrow 0$: Eddy momentum flux dominates

$\text{Ro} = \xi/f$: Local
Rossby number
 S : Eddy
momentum flux
divergence

Intermediate regimes

Eddy momentum flux

$$\psi(1 - \text{Ro}) \approx \frac{S}{f}$$

- $\text{Ro} \rightarrow 1$: Axisymmetric limit
- $\text{Ro} \rightarrow 0$: Eddy momentum flux dominates

$0 < \text{Ro} < 1$

There is no theory that captures how the Hadley circulation responds to climate changes in this intermediate range of Rossby numbers. -

[Levine and Schneider, 2011]

Annual mean and equinox HC are similar

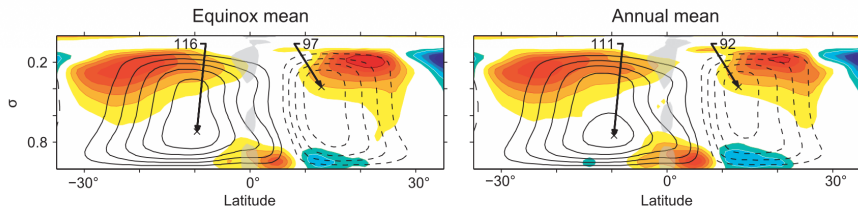
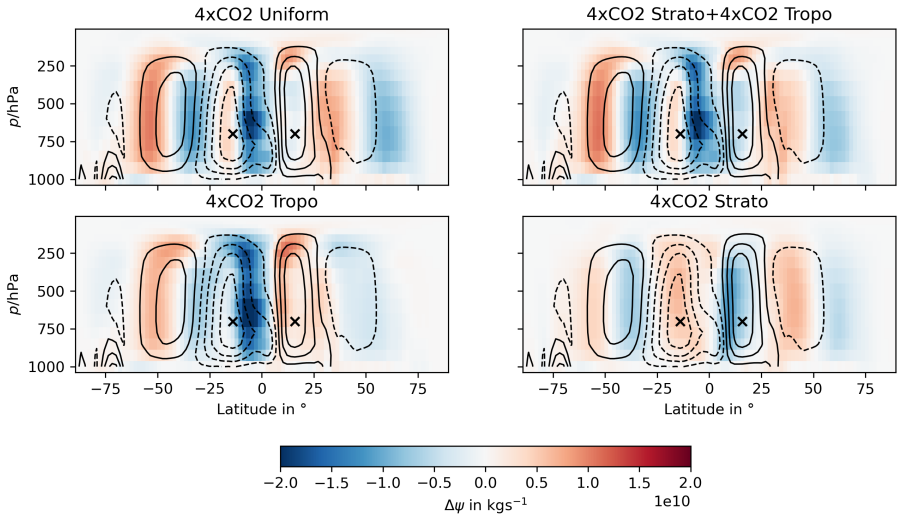
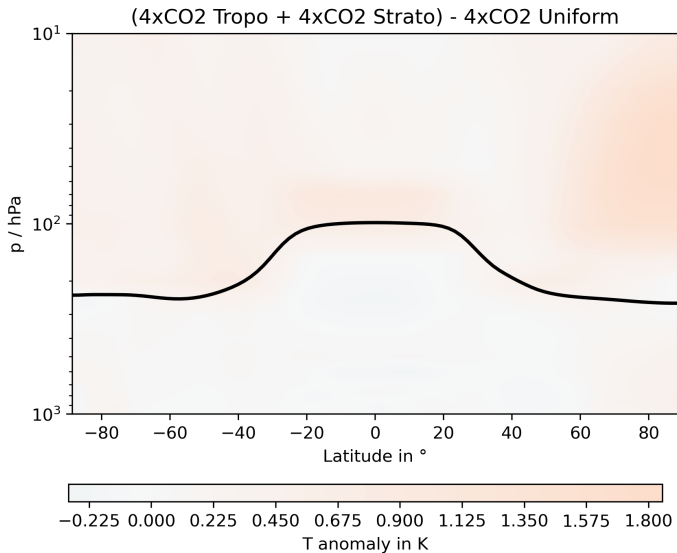


Figure: Local Rossby numbers in equinox and annual mean case using reanalysis data from [Levine and Schneider, 2011].

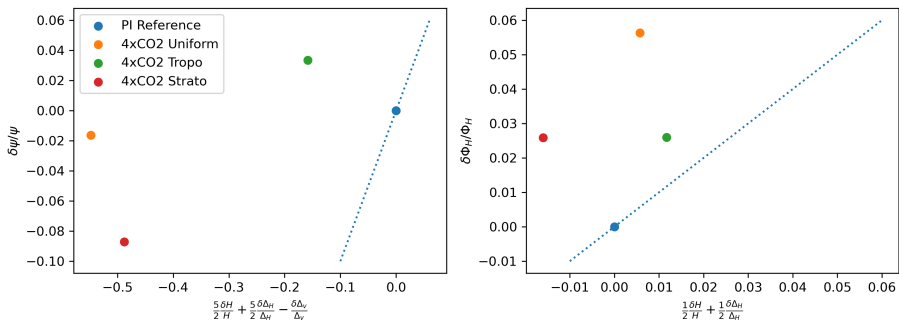
Zonally averaged circulation



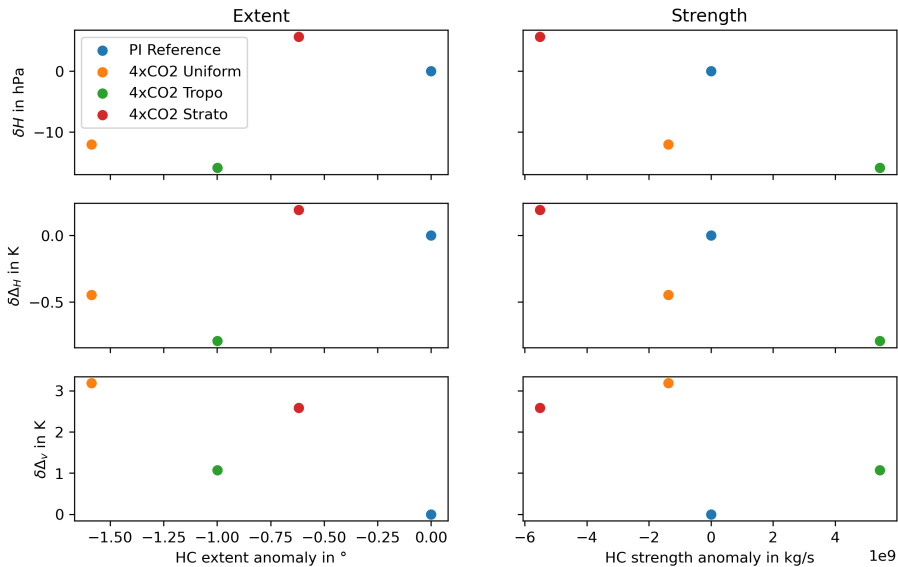
Is the decomposition valid?



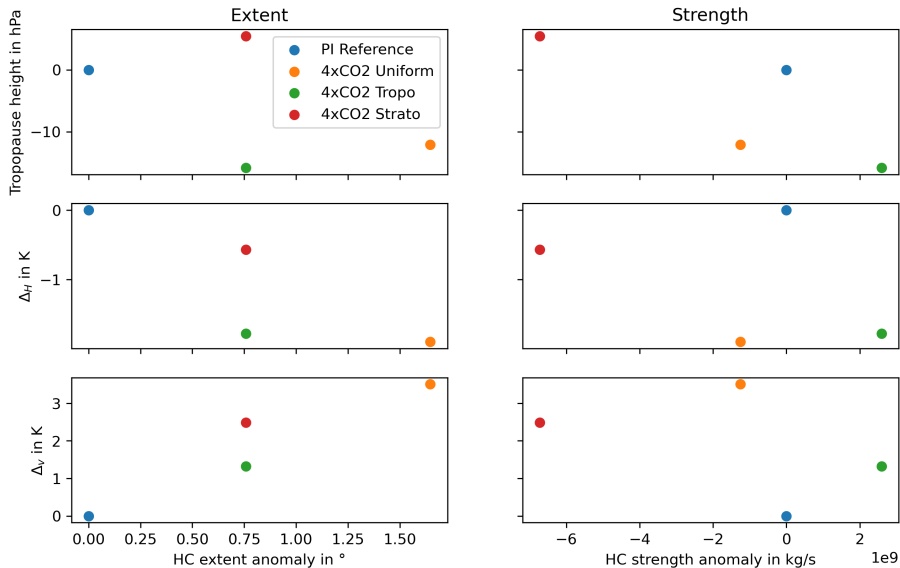
Northern hemisphere



Southern Hemisphere

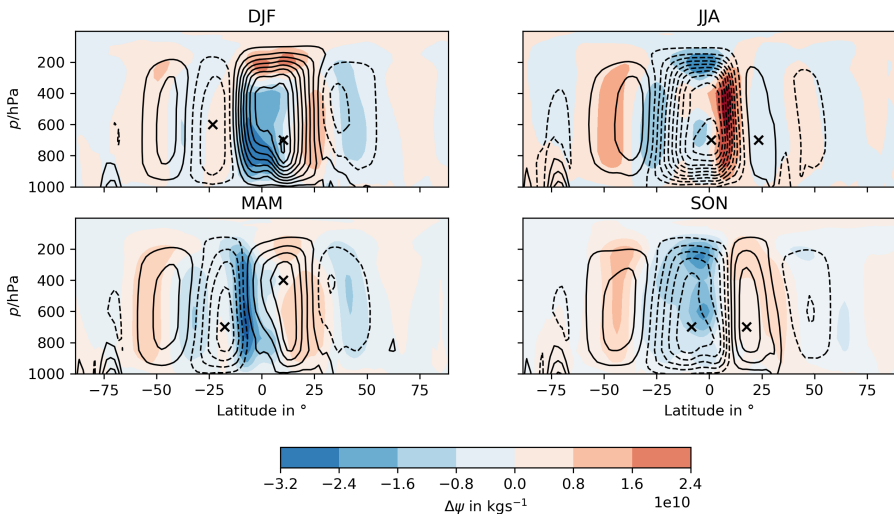


Northern Hemisphere

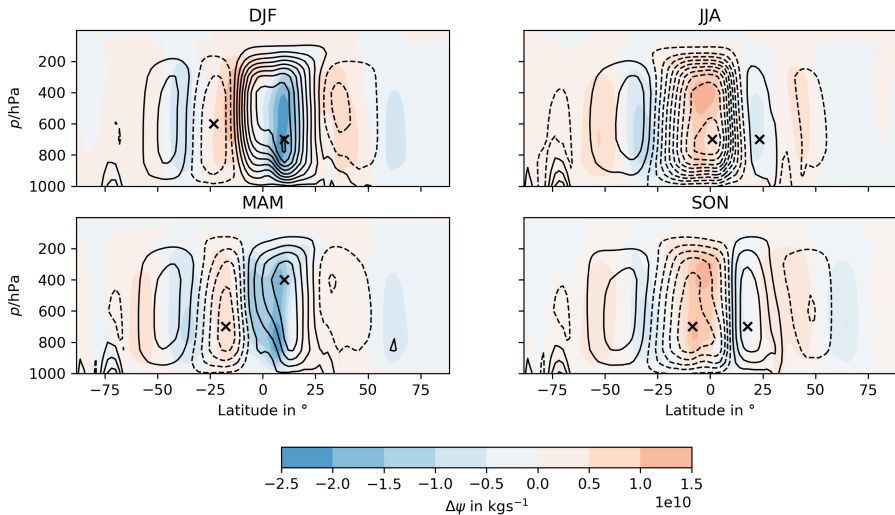


Winter HC is close to angular momentum conserving limit

4XCO₂ Tropo



4xCO2 Strato



4XCO2 Uniform

